

Effects of captivity on the cloacal microbiome of Dark-eyed Juncos (*Junco hyemalis*)

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INTRODUCTION

Background

- Microbiome (subset of all microorganisms associated with a host) contributes to overall health of the host
- Since microbiome can affect host phenotype, microbiome can play a role in host evolution
- Microbiome can be acquired through vertical transmission, the environment, and/or social contact
- Microbes in the uropygial gland produce volatile compounds detectable by other birds

Research Question: How does captivity affect the microbiome?

- Alpha diversity was lower in females (not significant) and in the regulatory subspecies ($P = 0.02$)
- Alpha diversity decreased over time in captivity
- Community composition was different between regulatory and resident birds ($P = 0.007$) and between the various time points of the experiment
- Shannon's diversity decreased over time in the experimental group ($P = 0.007$)
- Microbial composition was weakly correlated ($r = 0.148$, $P = 0.001$) with percent of abundance

Significance

- Understanding the microbiome of animals provides insight into how these organisms alter phenotypes, potentially changing fitness
- Learning how captivity can affect the microbiome is important for designing experiments to account for these alterations and their downstream effects on the health of the host organism

Goal of This Study

- Identify how sex, subspecies, and captivity lead to changes in the cloacal microbiome of dark-eyed juncos

METHODS

Animal Study

- Two subspecies of dark-eyed juncos (*Junco hyemalis*) subspecies "resident" and *J. h. hyemalis* "regulatory" were captured in late March 2019 at Mountain Lake Biological Station (MLBS) in Perkasie, PA
- Birds were housed in outdoor aviary at MLBS; fed a commercial diet of millet seeds and redworms
- Birds were housed and moved according to the protocol shown in Figure 1
- The cloaca was sampled with a swabbed cotton swab



Figure 1. Experimental design of juncos. The diagram shows the capture, housing, and sampling of regulatory and resident juncos. Each box represents a sex and subspecies (regulatory may perform regulatory tasks). Numbers in each box indicate the quantity of birds per time or per type. All pairs were housed separately. Figure 1 is from Whittaker et al. 2020.

DATA ANALYSIS

- DATA extracted from studies using QIAGEN Divergence PowerTool (Divergence PowerTool)
- Sequences were aligned using Geneious
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Statistical Analysis

- Paired-end, short-read reads were processed using DADA2 v1.26.0 in R
- Forward reads were truncated at 250bp; reverse at 200bp
- Taxonomy assigned using SILVA v138.1 with species level information
- Processed reads and taxonomy information passed to phyloseq v1.40.1
- Phyloseq is a separate measures tool using brd (you added using brd as random variable)
- Phyloseq used to remove rare taxa, calculate alpha diversity metrics, and calculate Bray-Curtis distance
- Other packages used include Microbiome, vegan, v2.0-2, and ggplot2 v3.3.6

RESULTS

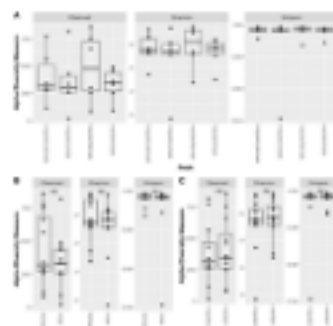


Figure 2. Alpha diversity metrics at time of capture. Shannon's diversity, Simpson's diversity, and Pielou's evenness for regulatory and resident birds. Significant differences were identified between regulatory and resident birds for Shannon's diversity and Simpson's diversity. Diversity was calculated using phyloseq v1.40.1.

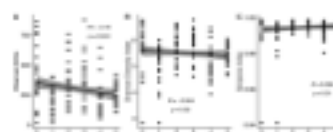


Figure 3. Alpha diversity decreases over time in captivity. Shannon's diversity, Simpson's diversity, and Pielou's evenness for regulatory and resident birds. Shannon's diversity decreased over time in captivity. Diversity was calculated using phyloseq v1.40.1.

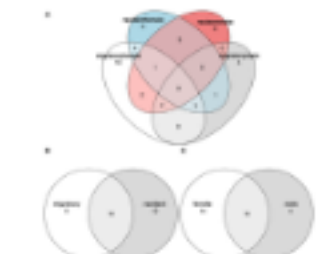


Figure 4. Number of taxa found in 10% of individuals at the time of capture. The Venn diagrams show the number of taxa found in 10% of individuals for regulatory and resident birds.

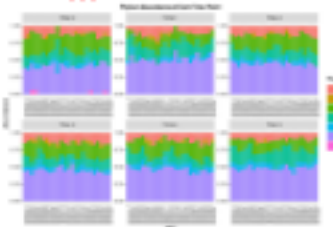


Figure 5. Beta diversity at each time point. A. Pielou's evenness and B. Simpson's diversity for regulatory and resident birds. Beta diversity was calculated using phyloseq v1.40.1.

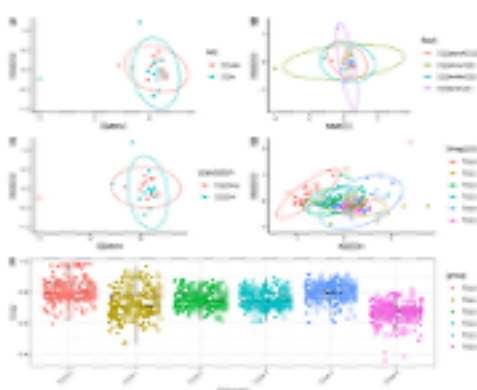


Figure 6. Beta diversity metrics at time of capture. Shannon's diversity, Simpson's diversity, and Pielou's evenness for regulatory and resident birds. Significant differences were identified between regulatory and resident birds for Shannon's diversity and Simpson's diversity. Diversity was calculated using phyloseq v1.40.1.

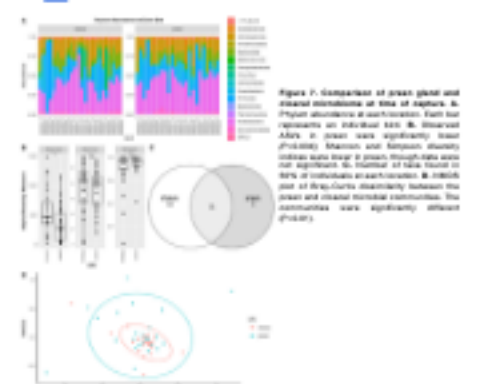


Figure 7. Comparison of mean and standard deviation of taxa. The box plots show the mean and standard deviation of taxa for regulatory and resident birds. Mean and standard deviation of taxa were higher in regulatory birds.

REFERENCES

- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.
- Whittaker, D. L., Peterson, N., Mando, J., Watkins, A., Van Laar, T., Slade, J., Theis, K., Greenberg, J. (2020). The cloacal microbiome of dark-eyed juncos (*Junco hyemalis*) is shaped by sex, subspecies, and captivity. *Microbiome*, 8, 1-10.

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Whittaker et al., 2020